Assignment6 – Learning

Given: May 27 Due: June 1

Problem 6.1 (Decision Trees)

Consider a word W chosen uniformly from $\{bad, bed, bend, pend, pad, ped\}$. You are allowed to ask the following questions about W:

A length of the word
B first letter of the word
C second letter of the word
D last letter of the word

1. Show that there is no decision tree for W of depth 2.

Solution: All questions have at most 2 possible answers, so a decision tree of depth 2 has at most 4 leaves. But we need at least 5 leaves to cover all options for W.

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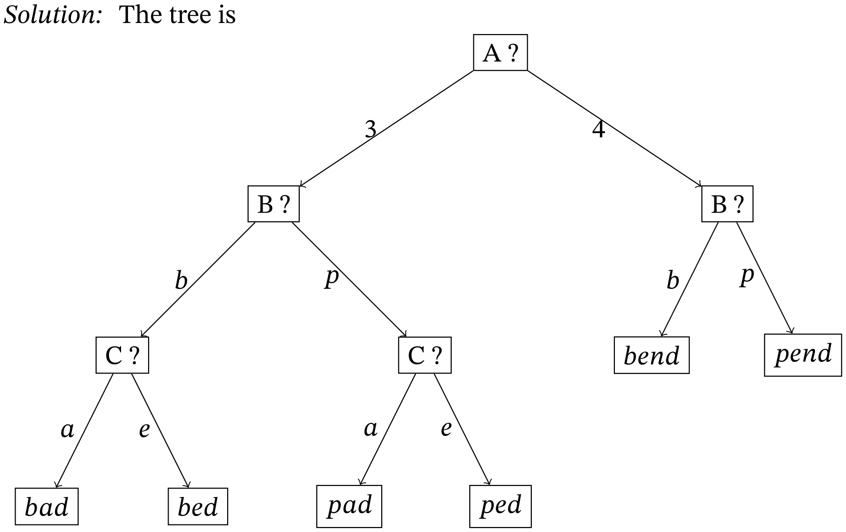
2. Draw the decision tree for W that arises from asking the questions in the order A,B,C,D. (Do not ask additional questions if the word can already be identified.)

Solution: The tree is



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3. Calculate the information gain for all 4 questions.

Solution: A, C: $-1/3\log_2 1/3 - 2/3\log_2 2/3$

B: $-1/2\log_2 1/2 - 1/2\log_2 1/2 = 1$
D: $-1\log_2 1 = 0$

4. Which question would the information gain algorithm ask first?

Solution: B

We write $Q \succ A$ if the set Q of attributes determines the attribute A, i.e., examples that agree on all attributes in Q also agree on A.

5. Give all minimal sets $Q \subseteq \{A, B, C, D\}$ of questions for which the determination $Q \succ W$ holds?

Solution: $\{A, B, C\}$

6. Give two words such that removing them from the choices for W makes the determination $\{A, B\} \succ W$ hold.

Solution: Any pair out of $\{bad, bed\} \times \{pad, ped\}$

Problem 6.2 (Decision Tree Learning in Python)

Implement the Decision Tree Learning algorithm (DTL) in Python using the files at `https://kwarc.info/teaching/AI/resources/AI2/dtl`.

Solution: See `https://kwarc.info/teaching/AI/resources/AI2/dtl`

Problem 6.3 (Loss)

Our goal is to find a linear approximation $h_a(x) = ax$ for the sequence of squares $0, 1, 4, 9, 16$ of the numbers $0, 1, 2, 3, 4$.

1. Model this situation as an inductive learning problem.

Solution: The inductive learning problem is $(\mathcal{H}, f)$ where

- the hypothesis space $\mathcal{H}$ is the set containing all functions $h_a(x) = ax$ with $\mathrm{dom}(h_a) = \{0, \ldots, 4\}$ for $a \in \mathbb{R}$
- the target function is $f(x) = x^2$ with $\mathrm{dom}(f) = \{0, 1, \ldots, 4\}$

2. Assuming all 5 possible examples are equally probable, compute the generalization loss using the squared error loss function.

Solution: Each example (x,x^2) has probability $1/5$. For each x, the loss is $L_2(x^2, ax) = (x^2 - ax)^2$. Thus for each $h(x) = ax$, we have

$$GenLoss(h_a) = \sum_{x=0}^{4}(x^2-ax)^2 \cdot 1/5 = ((1-a)^2+(4-2a)^2+(9-3a)^2+(16-4a)^2)/5 = (354-200a+30a^2)/5$$

3. Explain in about 2 sentences how you would continue to eventually determine that $h^*(x) = 10x/3$.

Solution: We need to find the a that minimizes the loss. The derivative of $GenLoss$ for a is $(60a - 200)/5$. So the minimum is at $a = 10/3$.

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4. What is the error rate of h^*?

Solution: The error rate is $4/5$ because $h^*(x) = 10x/3$ predicts 4 out of 5 examples incorrectly. (E.g., $h_a(x) = x$ would have better error rate $3/5$ despite having higher generalization loss.)

Problem 6.4 (Overfitting)

Explain what *overfitting* means and why we want to avoid it.

Solution: Overfitting is a modeling error that occurs when the chosen hypothesis is too closely fit to a sample set of data points. It picks an overly complex hypothesis that also explains idiosyncrasies and errors in the data. A simpler hypothesis that fits the data less exactly is often a better match for the underlying mechanisms.

Problem 6.5 (Competition (due September 15))

In this competition, you will implement an agent that explores the FAULumpus world. You will receive up to 2 percentage points of additional bonus for your agents.

Submission deadline for your agents is September 15. All further details will be posted in the respective matrix channel.